

11	A gas molecule of mass m moves with velocity v and collides elastically with a wall perpendicular to that direction. If the molecule makes N such collisions per second with the wall, what is the average force F exerted by the molecule on the wall?					
	A $F = mvN$	B $F = 2mvN$	C $F = \frac{mv^2}{2}$	D $F = \frac{mv}{N}$		
	Answer: B Each elastic collision with the wall causes a change in momentum: $\Delta p = -mv - (+mv) = -2mv$					

According to Newton's 3rd law of motion, the wall experiences a force equal in magnitude but opposite in direction. If the molecule hits the wall N times per second, the rate of change of momentum (i.e., force),

$$F = N \times \Delta p = N \times (2mv) \\ = 2mvN$$

12 A piece of metal of mass m , specific heat capacity c and temperature 20°C is placed into a liquid